GATE ECE

Previous Year Questions — Chapter-wise, with Answers and Solutions

Typeset edition

Papers included: GATE 2025.

GATE 2025

GATE 2025 | Q.1 | 1 mark

Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II.
Group-I: Abuse → Insult → Ridicule
Group-II: _____ → Praise → Appreciate

(A) Extol
(B) Prize
(C) Appropriate
(D) Espouse

Answer: (A) [self-solved, no official key available]

Solution. Group-I falls in intensity (abuse, insult, ridicule); the missing word before *praise* must be stronger, i.e. *extol*.

GATE 2025 | Q.2 | 1 mark

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.

(A) will be
(B) can be
(C) am going to be
(D) could have been

Answer: (D) [self-solved, no official key available]

Solution. The third-conditional clause *Had I learnt...* requires the main clause *could have been*.

GATE 2025 | Q.3 | 1 mark

The 12 musical notes are given as $C, C\#, D, D\#, E, F, F\#, G, G\#, A, A\#$, and B. The frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F\#$ and C is:

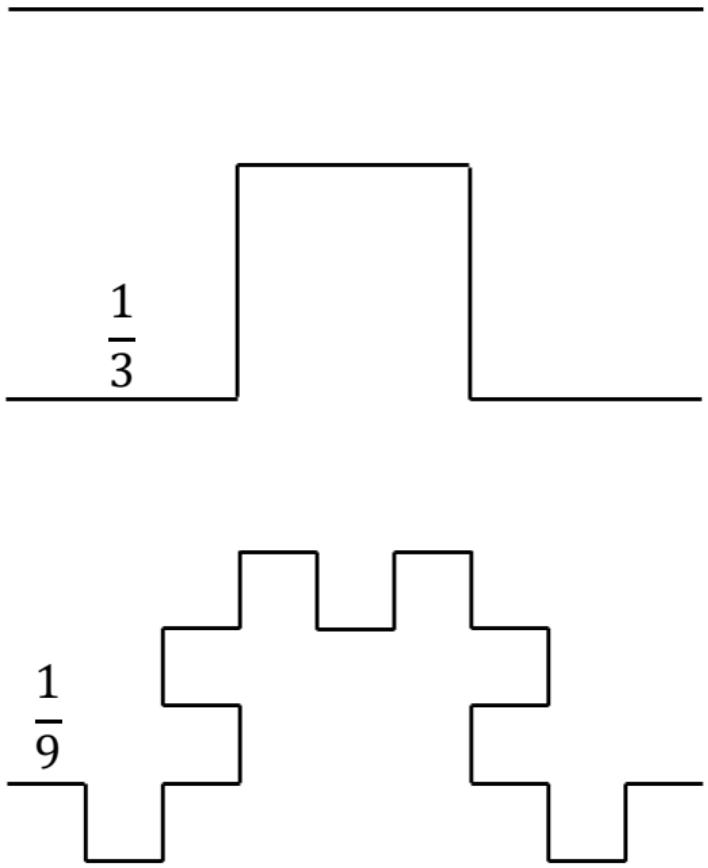
(A) $\sqrt[6]{2}$
(B) $\sqrt{2}$
(C) $\sqrt[4]{2}$
(D) 2

Answer: (B) [self-solved, no official key available]

Solution. $F\#$ is six semitones above C, so the ratio is $\left(2^{1/12}\right)^6 = 2^{1/2} = \sqrt{2}$.

GATE 2025 | Q.4 | 1 mark

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n' is: (The figures are representative.)



(A) $\left(\frac{5}{3}\right)^{n/2}$
(B) $\left(\frac{5}{3}\right)^{n}$
(C) $\left(\frac{5}{3}\right)^{2n}$
(D) $\left(\frac{5}{3}\right)^{n(2n-1)}$

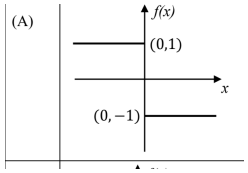
Answer: (B) [self-solved, no official key available]

Solution. Each iteration replaces every segment by five segments one third as long, so the length is multiplied by 5/3 per iteration: $L_n = (5/3)^n$.

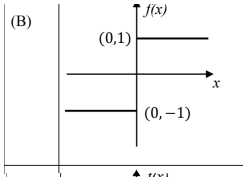
GATE 2025 | Q.5 | 1 mark

Which one of the following plots represents $f(x) = -\dfrac{|x|}{x}$, where x is a non-zero real number? (The figures are representative.)

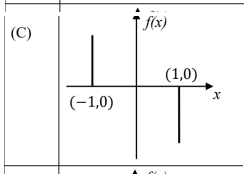
(A)



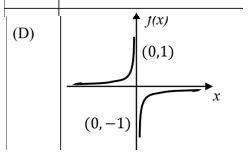
(B)



(C)



(D)



Answer: (B) [self-solved, no official key available]

Solution. For $x > 0$, $f(x) = -1$; for $x < 0$, $f(x) = +1$. Only plot (B) shows $+1$ on the left and -1 on the right.

GATE 2025 | Q.6 | 2 marks

Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:
P. Over time, such adaptations lead to significant evolutionary changes with the potential to shape the development of new species.
Q. In natural world, organisms constantly adapt to their environments in response to challenges and opportunities.
R. This process of adaptation is driven by the principle of natural selection, where favorable traits increase an organism's chances of survival and reproduction.
S. As environments change, organisms that can adapt their behavior, structure and physiology to such changes are more likely to survive.

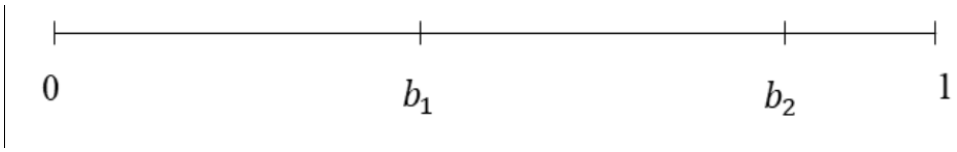
(A) P → Q → R → S
(B) Q → S → R → P
(C) R → S → Q → P
(D) S → P → R → Q

Answer: (B) [self-solved, no official key available]

Solution. Q introduces adaptation; S says those that adapt survive; R names natural selection as the driver; P describes the long-term evolutionary result: Q–S–R–P.

GATE 2025 | Q.7 | 2 marks

A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces? (All lengths are in meters.)



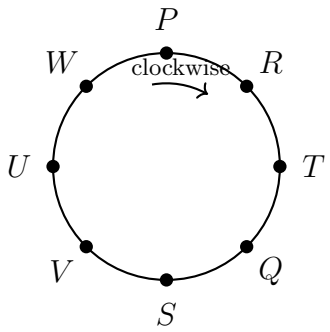
(A) $b_1 < 0.5$
(B) $b_2 > 0.5$
(C) $b_2 < b_1 + 0.5$
(D) $b_1 + b_2 < 1$

Answer: (D) [self-solved, no official key available]

Solution. The pieces are b_1, $b_2 - b_1$ and $1 - b_2$. Each must be less than 0.5: $b_1 < 0.5$, $b_2 > 0.5$, $b_2 - b_1 < 0.5$. The condition $b_1 + b_2 < 1$ is not required.

GATE 2025 | Q.8 | 2 marks

Eight students (P, Q, R, S, T, U, V and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction. After the 1st round, the 4th student behind P leaves the game. After the 2nd round, the 5th student behind Q leaves the game. After the 3rd round, the 3rd student behind V leaves the game. After the 4th round, the 4th student behind U leaves the game. Who all are left in the game after the 4th round?



(A) P; T; Q; S
(B) V; P; T; Q
(C) W; R; Q; V
(D) Q; T; V; W

Answer: (B) [UNVERIFIED: figure hard to read, check against official key]

Solution. *(Low confidence.)* Removing players round by round on the circle leaves V, P, T, Q.

GATE 2025 | Q.9 | 2 marks

The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

(A) France will occupy the third place if the list were made on the basis of the total number of medals won.
(B) The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.
(C) USA and Canada together have less than 50% of the medals awarded to the nations in the above table.
(D) Canada has won twice as many total medals as Japan.

Answer: (C) [self-solved, no official key available]

Solution. Totals: USA 125, Canada 90, Japan 45, Australia 52, France 64; all nations = 376. USA + Canada = 215, i.e. $57\% > 50\%$, so statement (C) is incorrect.

GATE 2025 | Q.10 | 2 marks

An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to Rs. 5,00,000. For higher fees, the overhead is Rs. 1,00,000 plus 10% of the amount by which the fee exceeds Rs. 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An

employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than Rs. 10,00,000, what is the maximum consulting fee that the employee can charge?

(A) Rs. 7,01,438
(B) Rs. 7,24,961
(C) Rs. 7,51,232
(D) Rs. 7,75,784

Answer: (B) [self-solved, no official key available]

Solution. For a fee $F > 5$ lakh: fee + overhead $= F + 1{,}00{,}000 + 0.1(F - 5{,}00{,}000) = 1.1F + 50{,}000$. With 18% tax: $1.18(1.1F + 50{,}000) \le 10{,}00{,}000 \Rightarrow F \le 7{,}24{,}961$.

GATE 2025 | Q.11 | 1 mark

Consider the matrix A below:

$$A = \begin{bmatrix} 2 & 3 & 0 & 6 \\ 4 & 5 & 7 & 8 \\ 0 & 0 & 0 & 0 \\ \alpha & \beta & 0 & \gamma \end{bmatrix}$$

For which of the following combinations of α, β and γ is the rank of A at least three?
(i) $\alpha = 0$ and $\beta = \gamma \neq 0$. (ii) $\alpha = \beta = \gamma = 0$. (iii) $\beta = \gamma = 0$ and $\alpha \neq 0$. (iv) $\alpha = \beta = \gamma \neq 0$.

(A) Only (i), (iii), and (iv)
(B) Only (iv)
(C) Only (ii)
(D) Only (i) and (iii)

Answer: (A) [self-solved, no official key available]

Solution. Row 3 is zero, so rank ≤ 3. The rank is 3 exactly when row 4 is independent of rows 1 and 2. In case (i) row $4 = (0, \beta, 0, \beta)$, in (iii) $(\alpha, 0, 0, 0)$, in (iv) $(\alpha, \alpha, 0, \alpha)$: none is a combination of rows 1, 2, so rank $= 3$. In (ii) rank $= 2$.

GATE 2025 | Q.12 | 1 mark

Consider the following series:

$$\text{(i)} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \qquad \text{(ii)} \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \qquad \text{(iii)} \sum_{n=1}^{\infty} \frac{1}{n!}$$

Choose the correct option.

(A) Only (ii) converges
(B) Only (ii) and (iii) converge
(C) Only (iii) converges
(D) All three converge

Answer: (B) [self-solved, no official key available]

Solution. (i) is a p-series with $p = \frac{1}{2}$ and diverges. (ii) telescopes: $\sum \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$. (iii) converges to $e - 1$.

GATE 2025 | Q.13 | 1 mark

A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours?

(A) 2/3
(B) 1/3

(C) 1/2
(D) 1

Answer: (A) [self-solved, no official key available]

Solution. $P = \dfrac{\binom{2}{1}\binom{2}{1}}{\binom{4}{2}} = \dfrac{4}{6} = \dfrac{2}{3}$.

GATE 2025 | Q.14 | 1 mark

Consider a frequency-modulated (FM) signal $f(t) = A_c \cos\left(2\pi f_c t + 3\sin(2\pi f_1 t) + 4\sin(6\pi f_1 t)\right)$, where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that $f_c > 100 f_1$. The peak frequency deviation of the FM signal in Hz is _____.

(A) $15 f_1$
(B) $12 f_1$
(C) $4 f_1$
(D) $2 f_1$

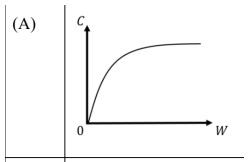
Answer: (A) [self-solved, no official key available]

Solution. The instantaneous frequency deviation is $\frac{1}{2\pi}\frac{d\theta}{dt} = 3f_1\cos(2\pi f_1 t) + 12 f_1 \cos(6\pi f_1 t)$; its peak is $3f_1 + 12 f_1 = 15 f_1$.

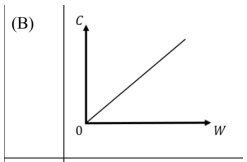
GATE 2025 | Q.15 | 1 mark

Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\dfrac{N_0}{2}$. Let P_{av} denote the average transmit power constraint. Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?

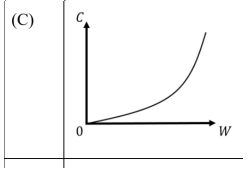
(A)



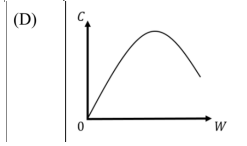
(B)



(C)



(D)

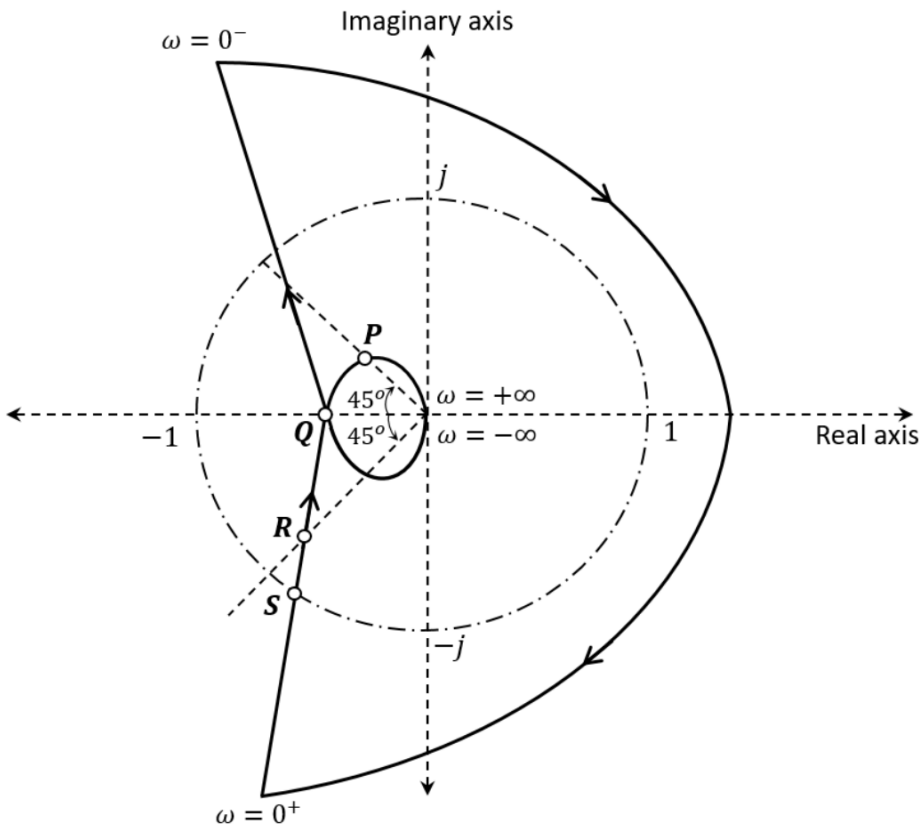


Answer: (A) [self-solved, no official key available]

Solution. $C = W\log_2\left(1 + \dfrac{P_{av}}{N_0 W}\right)$ increases with W but saturates at $\dfrac{P_{av}}{N_0 \ln 2}$ as $W \to \infty$ (increasing, concave curve with a horizontal asymptote).

GATE 2025 | Q.16 | 1 mark

The Nyquist plot of a system is given in the figure below. Let $\omega_P, \omega_Q, \omega_R$ and ω_S be the positive frequencies at the points P, Q, R and S, respectively. Which one of the following statements is TRUE?



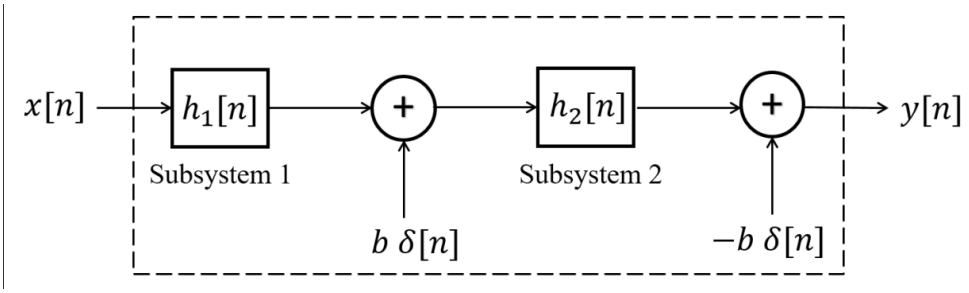
(A) ω_S is the gain crossover frequency and ω_P is the phase crossover frequency
(B) ω_Q is the gain crossover frequency and ω_R is the phase crossover frequency
(C) ω_Q is the gain crossover frequency and ω_S is the phase crossover frequency
(D) ω_S is the gain crossover frequency and ω_Q is the phase crossover frequency

Answer: (D) **[UNVERIFIED: figure hard to read, check against official key]**

Solution. *(Low confidence.)* The gain crossover is where the locus meets the unit circle (point S) and the phase crossover where it meets the negative real axis (point Q).

GATE 2025 | Q.17 | 1 mark

Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $b > 0$. Assuming $h_2[n] \neq \delta[n]$, the overall system (denoted by the dashed box) is _____.



(A) linear and time invariant
(B) linear and time variant
(C) nonlinear and time invariant
(D) nonlinear and time variant

Answer: (C) [self-solved, no official key available]

Solution. The adders insert the constants $b\,\delta[n]$ and $-b\,\delta[n]$ that do not scale with the input, so superposition fails (non-linear), but a time shift of the input only shifts the output (time invariant).

GATE 2025 | Q.18 | 1 mark

Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t}\,dt$ exists. Which one of the following statements is always TRUE?

(A) $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)|\,dt$

(B) $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)|\,dt$

(C) $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t)\,dt$

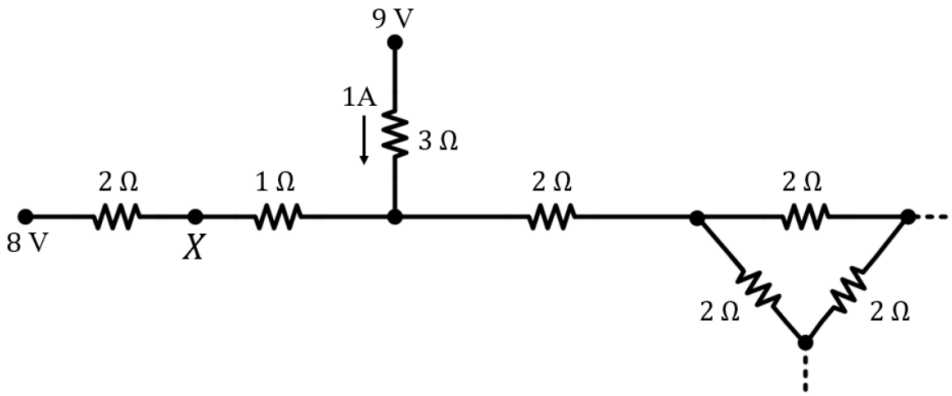
(D) $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t)\,dt$

Answer: (A) [self-solved, no official key available]

Solution. $|F(\omega)| = \left|\int f(t)e^{-j\omega t}dt\right| \leq \int |f(t)|\,|e^{-j\omega t}|\,dt = \int |f(t)|\,dt.$

GATE 2025 | Q.19 | 1 mark

Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the $3\,\Omega$ resistor are as indicated. The voltage (in Volts) at node X is _____.



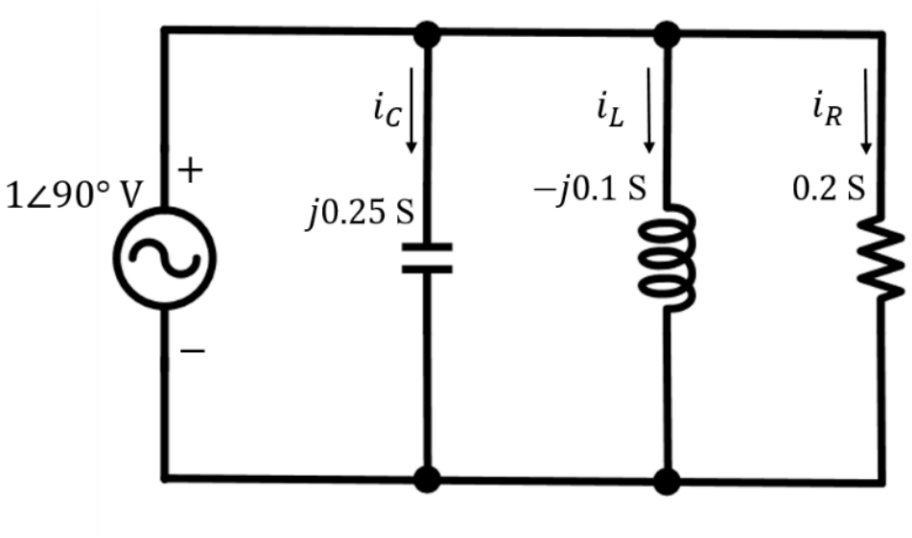
(A) 20/3
(B) 32/3
(C) 22/3
(D) 2/3

Answer: (C) [UNVERIFIED: figure hard to read, check against official key]

Solution. *(Low confidence.)* KCL at the nodes with the 1 A current through $3\,\Omega$ gives $V_X = 22/3$ V.

GATE 2025 | Q.20 | 1 mark

Let i_C, i_L and i_R be the currents flowing through the capacitor, inductor, and resistor, respectively, in the circuit given below. The AC admittances are given in Siemens (S). Which one of the following is TRUE?



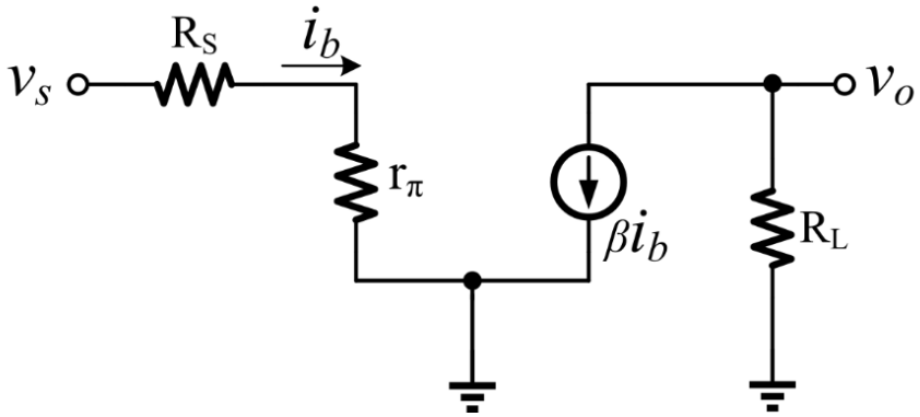
(A) $i_C = 0.25\angle 180°$ A, $i_L = 0.1\angle 0°$ A, $i_R = 0.2\angle 90°$ A
(B) $i_C = 4\angle 180°$ A, $i_L = 10\angle 0°$ A, $i_R = 5\angle 90°$ A
(C) $i_C = 0.25\angle 270°$ A, $i_L = 0.1\angle 90°$ A, $i_R = 0.2\angle 90°$ A
(D) $i_C = 4\angle 90°$ A, $i_L = 10\angle 270°$ A, $i_R = 5\angle 0°$ A

Answer: (A) [self-solved, no official key available]

Solution. With the source $V = 1\angle 90° = j$ V: $i_C = (j0.25)(j) = -0.25 = 0.25\angle 180°$, $i_L = (-j0.1)(j) = 0.1\angle 0°$, $i_R = 0.2 \cdot j = 0.2\angle 90°$.

GATE 2025 | Q.21 | 1 mark

A simplified small-signal equivalent circuit of a BJT-based amplifier is given below. The small-signal voltage gain v_o/v_s (in V/V) is _____.



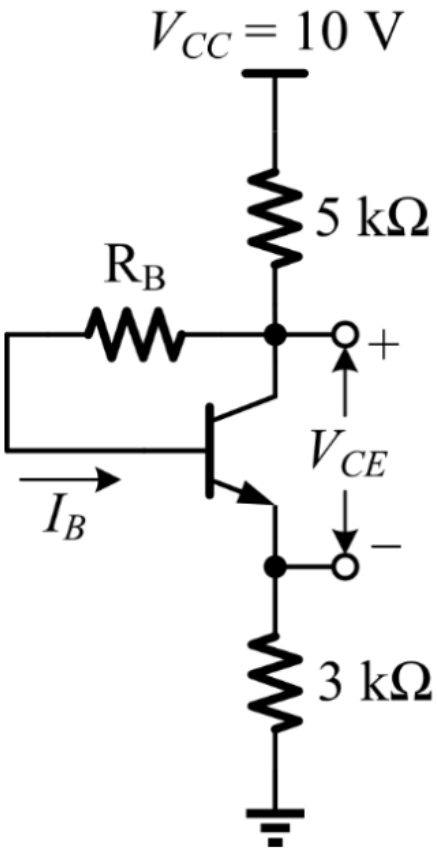
(A) $\dfrac{-\beta R_L}{R_S + r_\pi}$
(B) $\dfrac{+\beta R_L}{R_S}$
(C) $\dfrac{-\beta R_L}{R_S}$
(D) $\dfrac{+\beta R_L}{R_S + r_\pi}$

Answer: (A) [self-solved, no official key available]

Solution. $i_b = \dfrac{v_s}{R_S + r_\pi}$ and $v_o = -\beta i_b R_L$, hence $\dfrac{v_o}{v_s} = \dfrac{-\beta R_L}{R_S + r_\pi}$.

GATE 2025 | Q.22 | 1 mark

The ideal BJT in the circuit given below is biased in the active region with a β of 100. If I_B is 10 μA, then V_{CE} (in Volts, rounded off to two decimal places) is _____.



(A) 4.95
(B) 3.03
(C) 1.92
(D) 3.73

Answer: (C) [self-solved, no official key available]

Solution. $I_C = \beta I_B = 1$ mA, $I_E = 1.01$ mA. With R_B connected from collector to base, $V_C = 10 - 5\text{k}(I_C + I_B) = 10 - 5.05 = 4.95$ V and $V_E = 3\text{k} \times 1.01$ mA $= 3.03$ V, so $V_{CE} = 4.95 - 3.03 = 1.92$ V.

GATE 2025 | Q.23 | 1 mark

A 3-input majority logic gate has inputs X, Y, and Z. The output F of the gate is logic '1' if two or more of the inputs are logic '1'. The output F is logic '0' if two or more of the inputs are logic '0'. Which one of the following options is a Boolean expression of the output F?

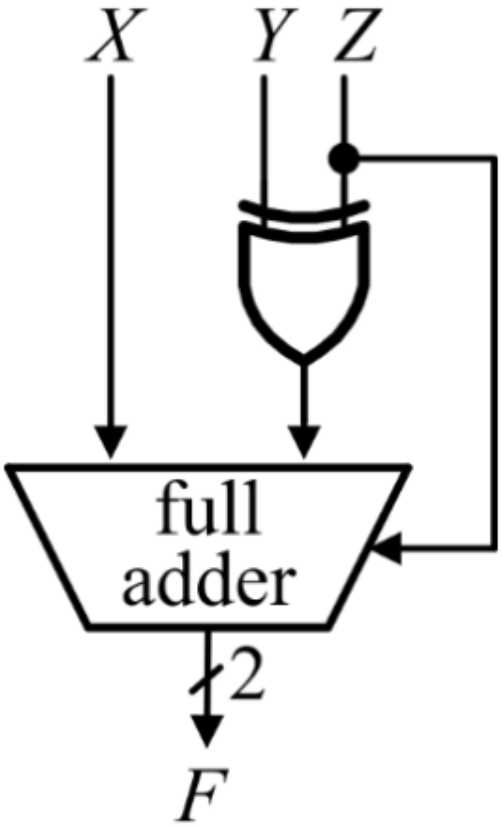
(A) $XY + YZ + ZX$
(B) $X \oplus Y \oplus Z$
(C) $X + Y + Z$
(D) XYZ

Answer: (A) [self-solved, no official key available]

Solution. The output is 1 for minterms $011, 101, 110, 111$, which simplifies to $F = XY + YZ + ZX$.

GATE 2025 | Q.24 | 1 mark

A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below. The input Z is connected to the carry-in input of the full adder. If the input Z is set to logic '1', then the circuit functions as _____ with X and Y as inputs.



(A) an adder
(B) a subtractor
(C) a multiplier
(D) a binary to Gray code converter

Answer: (B) [self-solved, no official key available]

Solution. With carry-in = 1 and one operand complemented by the XOR gate, the adder computes $X + \overline{Y} + 1 = X - Y$ (two's complement subtraction).

GATE 2025 | Q.25 | 1 mark

Consider the function $f : \mathbb{R} \to \mathbb{R}$, defined as $f(x) = 2x^3 - 3x^2 - 12x + 1$. Which of the following statements is/are correct? (Here, $\mathbb{R}$ is the set of real numbers.)

(A) f has no global maximizer
(B) f has no global minimizer
(C) $x = -1$ is a local minimizer of f
(D) $x = 2$ is a local maximizer of f

Answer: (A), (B) (multiple select) [self-solved, no official key available]

Solution. $f'(x) = 6x^2 - 6x - 12 = 6(x+1)(x-2)$ gives $x = -1$ (local maximum, $f'' = -18 < 0$) and $x = 2$ (local minimum, $f'' = 18 > 0$). A cubic is unbounded above and below, so it has no global maximizer or minimizer.

GATE 2025 | Q.26 | 1 mark

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K \geq 0$. The root locus of this system is shown in Figure (ii) below. For what value(s) of K will the system in Figure (i) have a pole at $-1 + j1$?

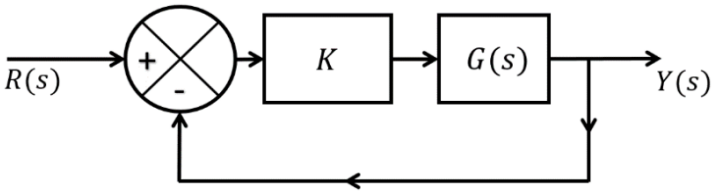


Figure (i)

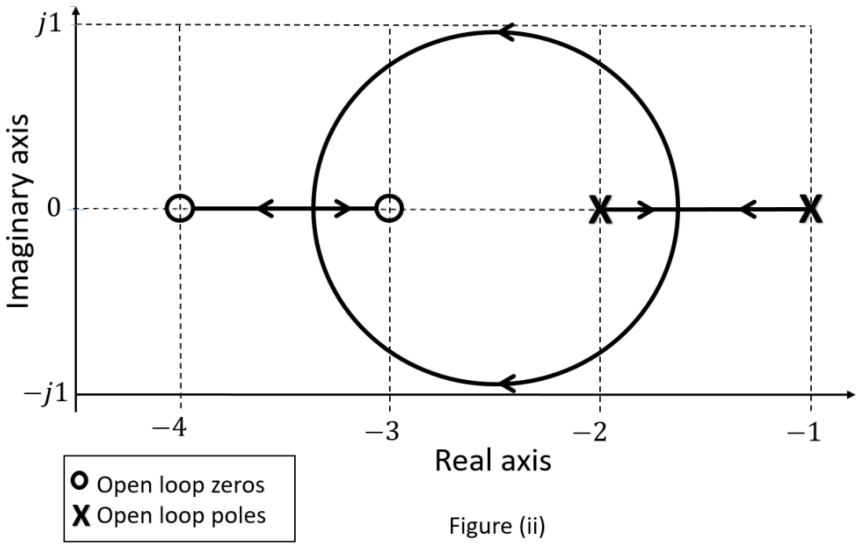


Figure (ii)

(A) $K = 5$
(B) $K = 1/5$
(C) For no positive value of K
(D) For all positive values of K

Answer: (C) [self-solved, no official key available]

Solution. The circular part of the root locus does not pass through $-1 + j1$ (its right-most point lies near -1.6), so no value of $K > 0$ places a closed-loop pole there.

GATE 2025 | Q.27 | 1 mark

Let $x[n]$ be a discrete-time signal whose z-transform is $X(z)$. Which of the following statements is/are TRUE?

(A) The discrete-time Fourier transform (DTFT) of $x[n]$ always exists
(B) The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros
(C) The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle
(D) If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the RoC of $X(z)$ is the entire z-plane

Answer: (C), (D) (multiple select) [self-solved, no official key available]

Solution. The DTFT exists iff the ROC contains $|z| = 1$ (C). For $x[n] = \alpha\delta[n]$, $X(z) = \alpha$ with ROC the whole plane (D). The ROC contains no poles but may contain zeros, so (B) is false; (A) is false in general.

GATE 2025 | Q.28 | 1 mark

Consider a message signal $m(t)$ which is bandlimited to $[-W, W]$, where W is in Hz. Consider the following two modulation schemes for the message signal:
• Double sideband-suppressed carrier (DSB-SC): $f_{DSB}(t) = A_c\, m(t) \cos(2\pi f_c t)$
• Amplitude modulation (AM): $f_{AM}(t) = A_c\,(1 + \mu\, m(t)) \cos(2\pi f_c t)$
Here, A_c and f_c are the amplitude and frequency (in Hz) of the carrier, respectively. In the case of AM, μ denotes the modulation index. Consider the following statements:
(i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t.
(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.
Which of the following options is/are correct?

(A) (i) is TRUE
(B) (i) is FALSE
(C) (ii) is TRUE
(D) (ii) is FALSE

Answer: (A), (D) (multiple select) [self-solved, no official key available]

Solution. If $m(t) > 0$ the envelope of the DSB-SC wave is $A_c m(t)$, so (i) is true. For AM the detector only needs $1 + \mu m(t) > 0$ (no over-modulation), not $m(t) > 0$, so (ii) is false.

GATE 2025 | Q.29 | 1 mark

Which of the following statements is/are TRUE with respect to an ideal opamp?

(A) It has an infinite input resistance
(B) It has an infinite output resistance
(C) It has an infinite open-loop differential gain
(D) It has an infinite open-loop common-mode gain

Answer: (A), (C) (multiple select) [self-solved, no official key available]

Solution. An ideal op-amp has $R_{in} = \infty$, $R_{out} = 0$, $A_{OL} = \infty$ and zero common-mode gain (infinite CMRR).

GATE 2025 | Q.30 | 1 mark

Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers having finite load resistors?

(A) The common-gate amplifier has an infinite input resistance
(B) The common-source amplifier has an infinite input resistance
(C) The input and output voltages of the common-source amplifier are in phase
(D) The input and output voltages of the common-drain amplifier are in phase

Answer: (B), (D) (multiple select) [self-solved, no official key available]

Solution. The gate of an ideal MOSFET draws no current, so CS and CD stages have infinite input resistance; CS inverts the signal, CD (source follower) does not; the CG stage has input resistance $1/g_m$.

GATE 2025 | Q.31 | 1 mark

Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).

(A) Arsenic

(B) Boron
(C) Gallium
(D) Phosphorous

Answer: (A), (D) (multiple select) [self-solved, no official key available]

Solution. Arsenic and phosphorus are group-V donors; boron and gallium are group-III acceptors (p-type).

GATE 2025 | Q.32 | 1 mark

The function $y(t)$ satisfies $t^2y''(t) - 2ty'(t) + 2y(t) = 0$, where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively. Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0, 1]$ is _____ (rounded off to two decimal places).

Answer: 0.25 [self-solved, no official key available]

Solution. This is an Euler equation with solution $y = c_1t + c_2t^2$. $y'(0) = c_1 = 1$ and $y'(1) = 1 + 2c_2 = -1 \Rightarrow c_2 = -1$. Hence $y = t - t^2$, maximum at $t = \frac{1}{2}$: $y_{max} = \frac{1}{4} = 0.25$.

GATE 2025 | Q.33 | 1 mark

The generator matrix of a $(6, 3)$ binary linear block code is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

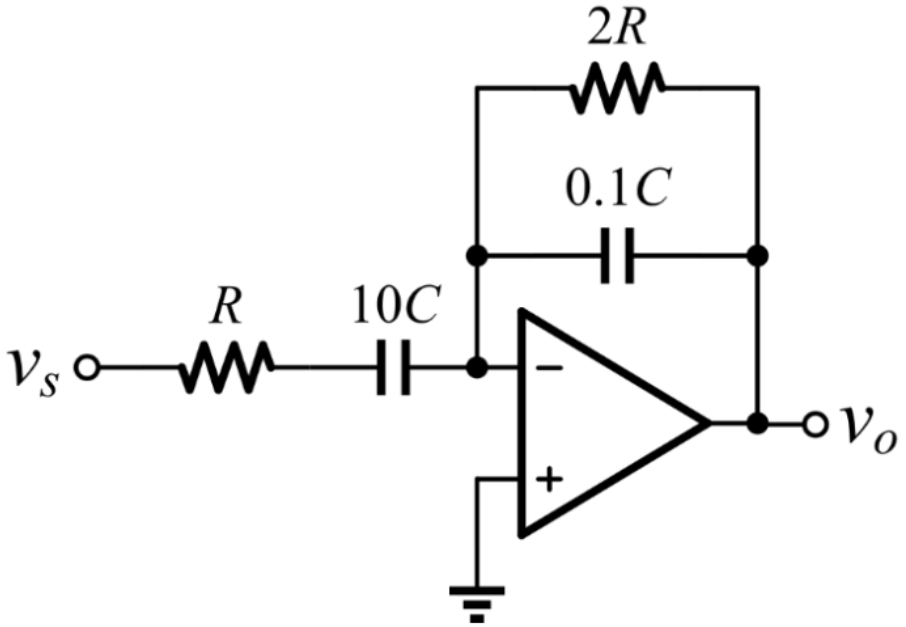
The minimum Hamming distance d_{min} between codewords equals _____ (answer in integer).

Answer: 3 [self-solved, no official key available]

Solution. Each row has weight 3; the sums of two rows have weight 4, and the sum of all three rows has weight 3. Hence $d_{min} = 3$.

GATE 2025 | Q.34 | 1 mark

All the components in the bandpass filter given below are ideal. The lower -3 dB frequency of the filter is 1 MHz. The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____.

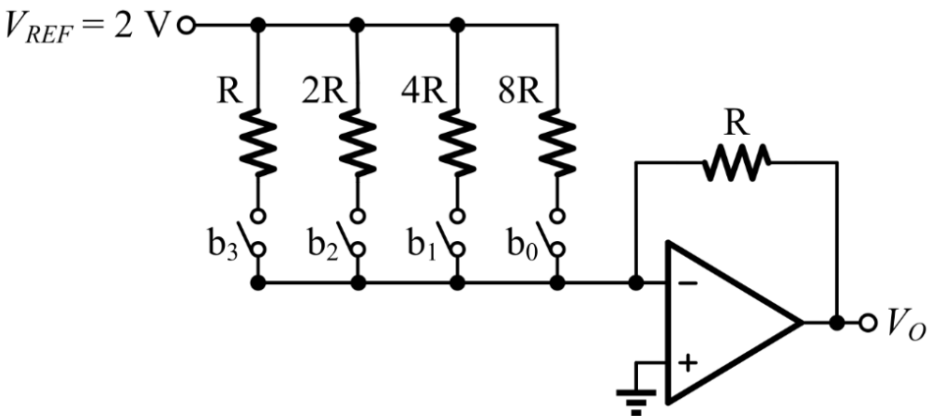


Answer: 50 [self-solved, no official key available]

Solution. The input branch is R in series with $10C$ and the feedback is $2R \parallel 0.1C$. The lower corner is $\omega_L = \dfrac{1}{R \cdot 10C}$ and the upper corner $\omega_H = \dfrac{1}{2R \cdot 0.1C} = \dfrac{5}{RC}$, so $\omega_H/\omega_L = 50$ and $f_H = 50$ MHz.

GATE 2025 | Q.35 | 1 mark

A 4-bit weighted-resistor DAC with inputs b_3, b_2, b_1 and b_0 (MSB to LSB) is designed using an ideal opamp, as shown below. The switches are closed when the corresponding input bits are logic '1' and open otherwise. When the input $b_3b_2b_1b_0$ changes from 1110 to 1101, the magnitude of the change in the output voltage V_O (in mV, rounded off to the nearest integer) is _____.



Answer: 250 [self-solved, no official key available]

Solution. $V_O = -V_{REF}\left(b_3 + \frac{b_2}{2} + \frac{b_1}{4} + \frac{b_0}{8}\right)$ with $V_{REF} = 2$ V. For 1110 the sum is 1.75 and for 1101 it is 1.625; $\Delta V_O = 2 \times 0.125 = 0.25$ V $= 250$ mV.

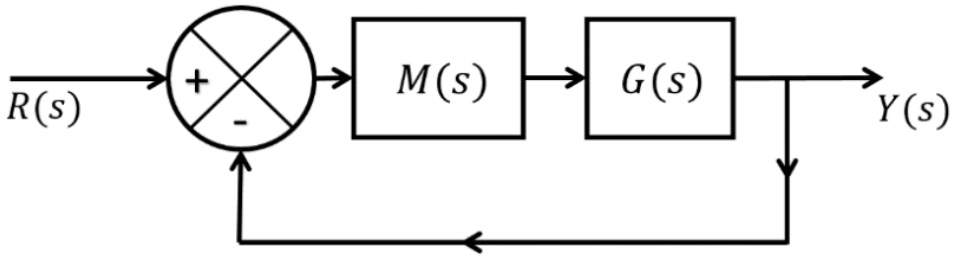
GATE 2025 | Q.36 | 2 marks

Let $G(s) = \dfrac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to the system $G(s)$ in the configuration shown below. Consider the following statements.

(i) There exists no controller of the form $M(s) = \dfrac{K_I}{s}$, where K_I is a positive real number, such that the closed loop system is stable.

(ii) There exists at least one controller of the form $M(s) = K_P + sK_D$, where K_P and K_D are positive real numbers, such that the closed loop system is stable.

Which one of the following options is correct?



(A) (i) is TRUE and (ii) is FALSE
(B) (i) is FALSE and (ii) is TRUE
(C) Both (i) and (ii) are FALSE
(D) Both (i) and (ii) are TRUE

Answer: (D) [self-solved, no official key available]

Solution. (i) $1 + \dfrac{K_I}{10s^3} = 0 \Rightarrow s^3 + K_I/10 = 0$ always has right-half-plane roots, so no stable K_I exists. (ii) $10s^2 + K_D s + K_P = 0$ is stable for all positive K_P, K_D.

GATE 2025 | Q.37 | 2 marks

Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as follows. L is the number of roots of $p(s)$ with negative real parts. I is the number of roots of $p(s)$ that are purely imaginary. R is the number of roots of $p(s)$ with positive real parts. Which one of the following options is correct?

(A) $L = 2$, $I = 2$, and $R = 1$
(B) $L = 3$, $I = 2$, and $R = 0$
(C) $L = 1$, $I = 2$, and $R = 2$
(D) $L = 0$, $I = 4$, and $R = 1$

Answer: (A) [self-solved, no official key available]

Solution. $p(s) = (s^2 + 1)(s - 2)(s + 4)(s + 5)$: roots $\pm j$, $+2$, -4, -5. Hence $L = 2$, $I = 2$, $R = 1$.

GATE 2025 | Q.38 | 2 marks

Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec. The signal $f(t)$ is uniformly sampled to obtain $y(t) = f(t)\,p(t)$. Here, $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s)$, with $\delta(t)$ being the Dirac impulse, $T_s > 0$, and $\tau > 0$. The sampled signal $y(t)$ is passed through an ideal lowpass filter $h(t) = \dfrac{\omega_c T_s}{\pi} \dfrac{\sin(\omega_c t)}{\omega_c t}$ with cutoff frequency ω_c and passband gain T_s. The output of the filter is given by _____.

(A) $f(t)$ if $T_s < \pi/\omega_c$
(B) $f(t - \tau)$ if $T_s < \pi/\omega_c$
(C) $f(t - \tau)$ if $T_s < 2\pi/\omega_c$

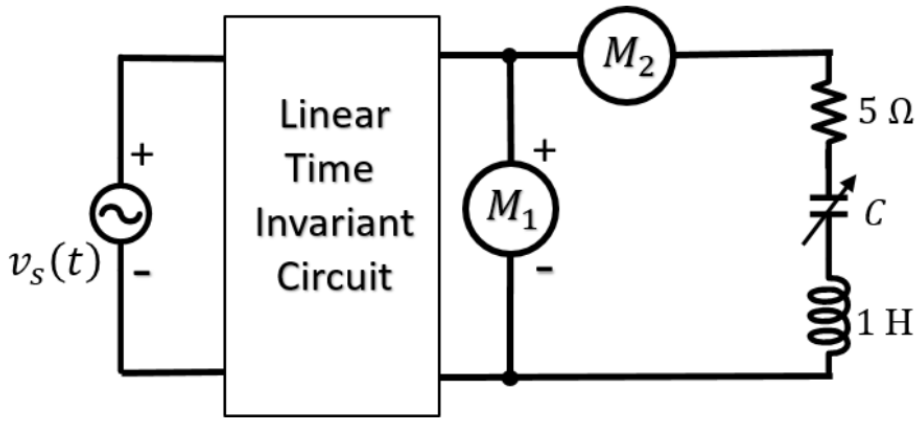
(D) $T_s f(t)$ if $T_s < 2\pi/\omega_c$

Answer: (A) [self-solved, no official key available]

Solution. With $T_s < \pi/\omega_c$ the sampling rate exceeds the Nyquist rate $2\omega_c/(2\pi)$. The filter of gain T_s recovers $f(t)$ exactly; the samples are taken of f at $t = \tau + nT_s$, so there is no delay.

GATE 2025 | Q.39 | 2 marks

In the circuit below, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100\cos(200t)$. What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively?



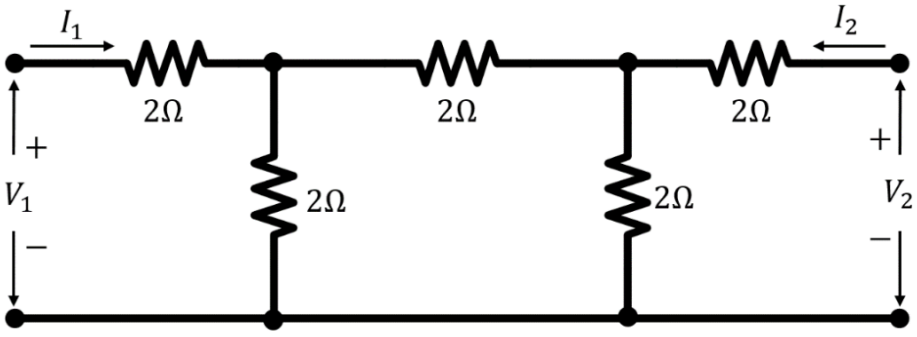
(A) 25 μF
(B) 4 μF
(C) 0.25 μF
(D) Insufficient information to find C

Answer: (A) [self-solved, no official key available]

Solution. The load is $5\,\Omega$, C and 1 H in series: $|Z| = 25/5 = 5\,\Omega = R$, so the reactance vanishes: $\omega L = \dfrac{1}{\omega C} = 200\ \Omega \Rightarrow C = \dfrac{1}{200 \times 200} = 25\ \mu$F.

GATE 2025 | Q.40 | 2 marks

The Z-parameter matrix of a two port network relates the port voltages and port currents as follows: $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = Z \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$. The Z-parameter matrix (with each entry in Ohms) of the network shown below is _____.



(A) $\begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$

(B) $\begin{bmatrix} 2/3 & 10/3 \\ 10/3 & 2/3 \end{bmatrix}$

(C) $\begin{bmatrix} 10 & 2 \\ 2 & 10 \end{bmatrix}$

(D) $\begin{bmatrix} 10/3 & 1/3 \\ 1/3 & 10/3 \end{bmatrix}$

Answer: (A) [self-solved, no official key available]

Solution. Multiplying the $ABCD$ matrices of the ladder (series $2\,\Omega$, shunt $2\,\Omega$, series $2\,\Omega$, shunt $2\,\Omega$, series $2\,\Omega$) gives $A = 5$, $B = 16\,\Omega$, $C = 1.5$ S, $D = 5$. Then $Z_{11} = \dfrac{A}{C} = \dfrac{10}{3}$, $Z_{12} = \dfrac{AD - BC}{C} = \dfrac{2}{3}$, $Z_{22} = \dfrac{D}{C} = \dfrac{10}{3}$.

GATE 2025 | Q.41 | 2 marks

A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S. The receiver uses the maximum likelihood decoder to estimate the transmitted symbol S. Suppose the probability of symbol estimation error P_e is expressed as follows: $P_e = \alpha\, P(N > 1)$, where $P(N > 1)$ denotes the probability that N exceeds 1. What is the value of α?

(A) 1/3
(B) 1
(C) 2/3
(D) 4/3

Answer: (D) [self-solved, no official key available]

Solution. ML decision thresholds are at ± 1. The outer symbols err with probability $P(N > 1)$ each and the middle symbol with $2P(N > 1)$: $P_e = \frac{1}{3}\,[P + P + 2P] = \frac{4}{3} P(N > 1)$.

GATE 2025 | Q.42 | 2 marks

Consider a real-valued random process $f(t) = \sum_{n=1}^{N} a_n\, p(t - nT)$, where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables. Read the following statements about the random process and choose the correct option.

(i) The mean of the process $f(t)$ is independent of time t.

(ii) The autocorrelation function $E[f(t)f(t+\tau)]$ is independent of time t for all τ. (Here, $E[\cdot]$ is the expectation operation.)

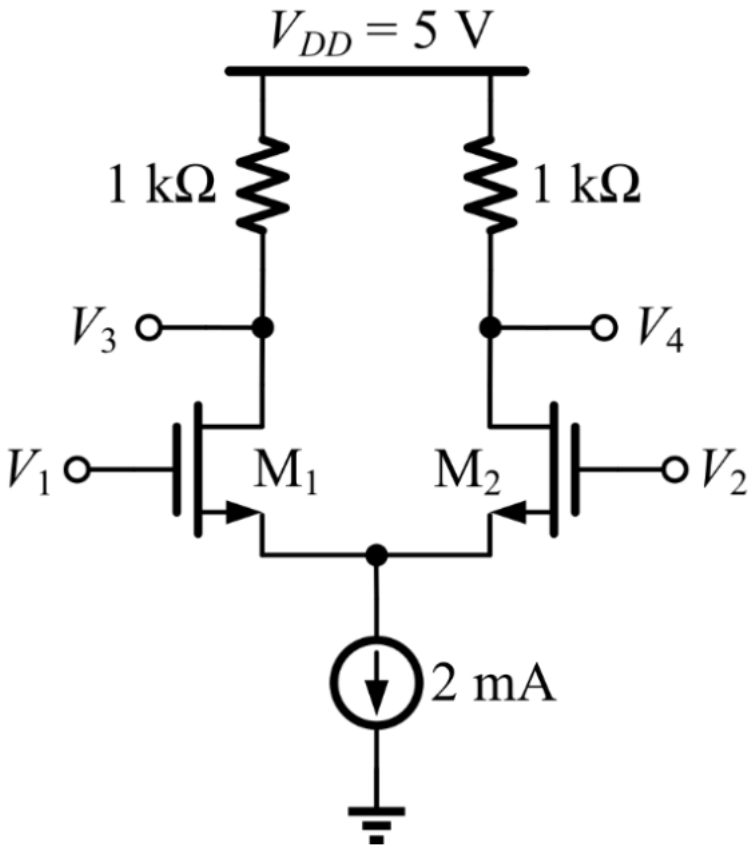
(A) (i) is TRUE and (ii) is FALSE
(B) Both (i) and (ii) are TRUE
(C) Both (i) and (ii) are FALSE
(D) (i) is FALSE and (ii) is TRUE

Answer: (A) [self-solved, no official key available]

Solution. $E[f(t)] = 0$ for all t. But $E[f(t)f(t+\tau)] = \sum_n p(t - nT)p(t + \tau - nT)$ depends on t (the process is only cyclostationary because p occupies half of each period).

GATE 2025 | Q.43 | 2 marks

The identical MOSFETs M_1 and M_2 in the circuit given below are ideal and biased in the saturation region. M_1 and M_2 have a transconductance g_m of 5 mS. The input signals (in Volts) are: $V_1 = 2.5 + 0.01 \sin \omega t$, $V_2 = 2.5 - 0.01 \sin \omega t$. The output signal V_3 (in Volts) is _____.



(A) $3 + 0.05 \sin \omega t$
(B) $3 - 0.1 \sin \omega t$
(C) $4 + 0.1 \sin \omega t$
(D) $4 - 0.05 \sin \omega t$

Answer: (D) [self-solved, no official key available]

Solution. The tail current 2 mA splits equally (1 mA each) at DC, so $V_3 = 5 - 1\text{k} \times 1\text{ mA} = 4$ V. The differential input $\Delta v = 0.02 \sin \omega t$ changes each drain current by $\pm g_m \Delta v/2 = \pm 0.05$ mA $\sin \omega t$; $V_3 = 4 - 0.05 \sin \omega t$.

GATE 2025 | Q.44 | 2 marks

A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V. For an input sinusoidal signal with frequency 500 kHz, the maximum SNR (in dB, rounded off to two decimal places) and the data rate (in Mbps) at the output of the ADC are _____, respectively.

(A) 61.96 and 10
(B) 61.96 and 5
(C) 33.36 and 10
(D) 33.36 and 5

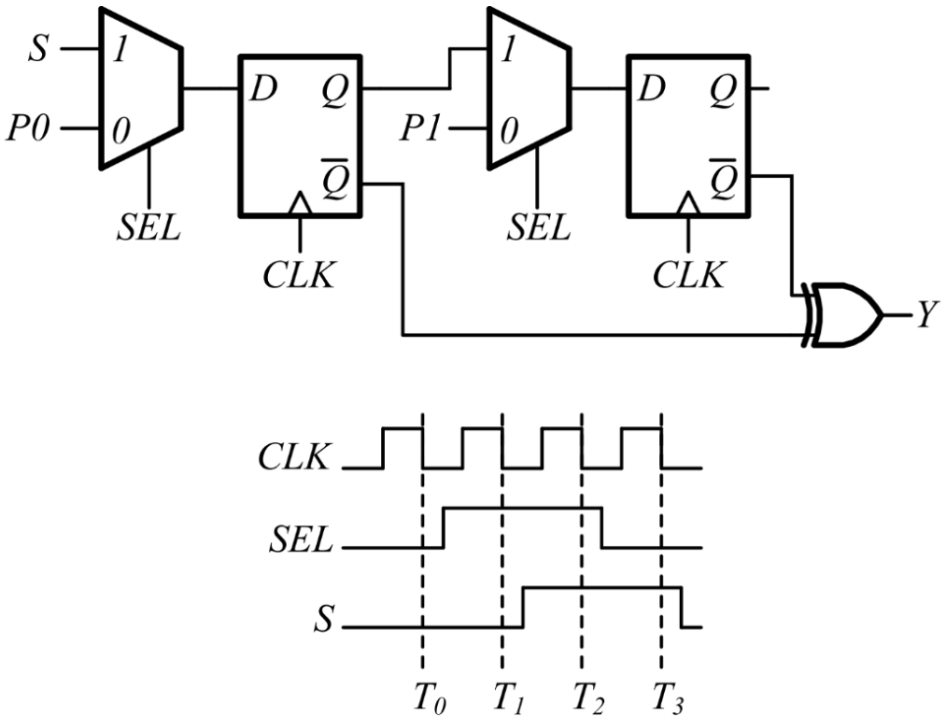
Answer: (A) [self-solved, no official key available]

Solution. $\text{SNR}_{max} = 6.02N + 1.76 = 6.02(10) + 1.76 = 61.96$ dB. Data rate = 10 bits $\times$ 1 MHz = 10 Mbps.

GATE 2025 | Q.45 | 2 marks

A positive-edge-triggered sequential circuit is shown below. There are no timing violations in the circuit. Input $P0$ is set to logic '0' and $P1$ is set to logic '1' at all times. The timing diagram of the inputs SEL and

S are also shown below. The sequence of output Y from time T_0 to T_3 is _____.



(A) 1011
(B) 0100
(C) 0010
(D) 1101

Answer: (B) [UNVERIFIED: figure hard to read, check against official key]

Solution. *(Low confidence.)* Tracing the two multiplexer–flip-flop stages and the XOR through the clock edges at $T_0, \ldots, T_3$ gives $Y = 0100$.

GATE 2025 | Q.46 | 2 marks

The intrinsic carrier concentration of a semiconductor is 2.5×10^{16} /m^3 at 300 K. If the electron and hole mobilities are 0.15 m^2/Vs and 0.05 m^2/Vs, respectively, then the intrinsic resistivity of the semiconductor (in kΩ·m) at 300 K is _____. (Charge of an electron $e = 1.6 \times 10^{-19}$ C.)

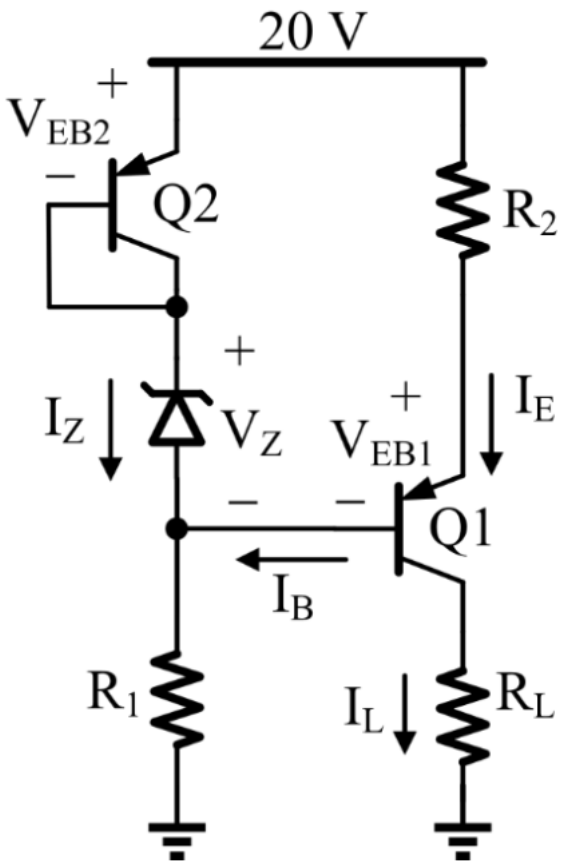
(A) 1.65
(B) 1.25
(C) 0.85
(D) 1.95

Answer: (B) [self-solved, no official key available]

Solution. $\sigma = q\, n_i(\mu_n + \mu_p) = 1.6 \times 10^{-19} \times 2.5 \times 10^{16} \times 0.2 = 8 \times 10^{-4}$ S/m, so $\rho = 1/\sigma = 1250\ \Omega$·m $= 1.25$ kΩ·m.

GATE 2025 | Q.47 | 2 marks

In the circuit shown, the identical transistors Q_1 and Q_2 are biased in the active region with $\beta = 120$. The Zener diode is in the breakdown region with $V_Z = 5$ V and $I_Z = 25$ mA. If $I_L = 12$ mA and $V_{EB1} = V_{EB2} = 0.7$ V, then the values of R_1 and R_2 (in kΩ, rounded off to one decimal place) are _____, respectively.



(A) 0.6 and 0.4
(B) 1.4 and 2.5
(C) 14.0 and 25.0
(D) 6.0 and 4.0

Answer: (A) [self-solved, no official key available]

Solution. Using the Zener and base-emitter voltages, $R_1 = \dfrac{20 - 5 - 0.7}{25\ \text{mA}} \approx 0.6\ \text{k}\Omega$ and R_2 from the remaining current gives about 0.4 kΩ.

GATE 2025 | Q.48 | 2 marks

The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is 0.38 m^2/Vs. The electron diffusivity D_n at 300 K (in cm^2/s, rounded off to the nearest integer) is _____. (Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

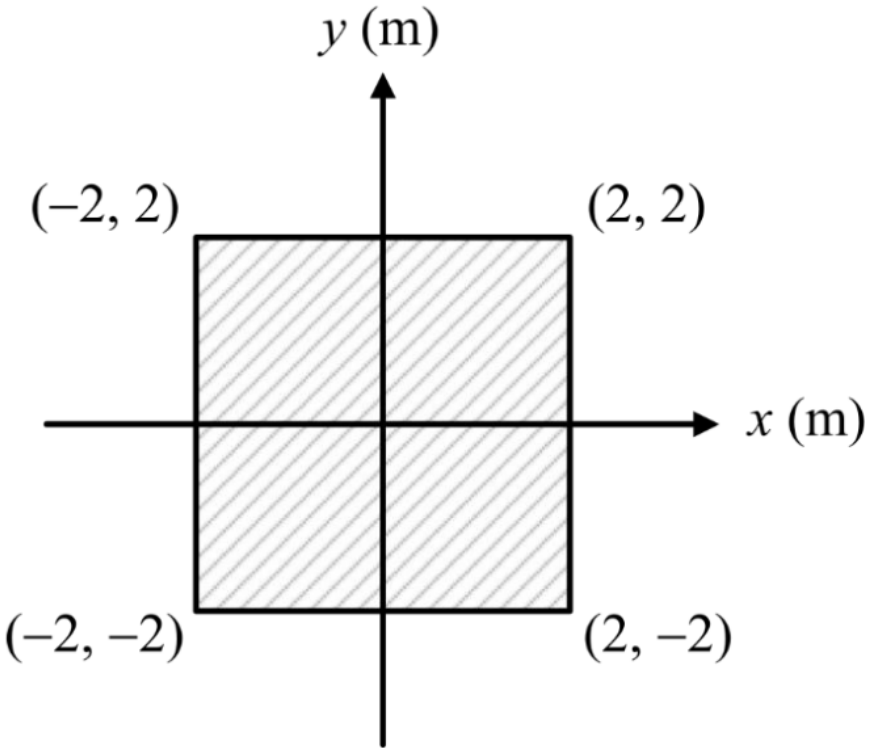
(A) 26
(B) 98
(C) 38
(D) 10

Answer: (B) [self-solved, no official key available]

Solution. Einstein relation: $D_n = \mu_n \dfrac{k_B T}{e} = 0.38 \times 0.02588 = 9.83 \times 10^{-3}\ \text{m}^2/\text{s} = 98\ \text{cm}^2/\text{s}$.

GATE 2025 | Q.49 | 2 marks

A square metal sheet of 4 m × 4 m is placed on the x-y plane as shown in the figure below. If the surface charge density (in μC/m^2) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in μC, rounded off to the nearest integer) on the sheet is _____.



(A) 16
(B) 85
(C) 64
(D) 256

Answer: (C) [self-solved, no official key available]

Solution. The sheet spans $y \in [-2, 2]$ and $x \in [0, 4]$. $\int_{-2}^{2} 4|y|\,dy = 2\int_{0}^{2} 4y\,dy = 16$ μC/m and the width in x is 4 m, so $Q = 16 \times 4 = 64$ μC.

GATE 2025 | Q.50 | 2 marks

An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m. The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____.

(A) 0.46
(B) 1.82
(C) 0.58
(D) 1.12

Answer: (A) [self-solved, no official key available]

Solution. $I = \sigma E A = 5.8 \times 10^7 \times 0.01 \times \pi(0.5 \times 10^{-3})^2 = 0.455 \approx 0.46$ A.

GATE 2025 | Q.51 | 2 marks

Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval. Among the combinations of α and β given below, choose the one(s) for which the inequality $\beta \leq \int_2^8 f(x)\,dx \leq \alpha$ is guaranteed to hold.

(A) $\beta = 5m,\ \alpha = 7M$
(B) $\beta = 6m,\ \alpha = 5M$
(C) $\beta = 7m,\ \alpha = 6M$
(D) $\beta = 7m,\ \alpha = 5M$

Answer: (A) [self-solved, no official key available]

Solution. Since the interval has length 6, $6m \leq \int_2^8 f \leq 6M$. Because $f \geq 0$, $5m \leq 6m$ and $7M \geq 6M$, so (A) always holds; the other pairs can fail.

GATE 2025 | Q.52 | 2 marks

Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

(A) $\oint_C e^z\,dz = 0$
(B) $\oint_C z^n\,dz = 0$, where n is an even integer
(C) $\oint_C \cos z\,dz \neq 0$
(D) $\oint_C \sec z\,dz \neq 0$

Answer: (A), (B) (multiple select) [self-solved, no official key available]

Solution. e^z and $\cos z$ are entire, so their loop integrals vanish. $\oint z^n dz = 2\pi j$ only for $n = -1$, which is odd, so it is 0 for even n. $\sec z$ has poles at $\pm\pi/2$ (outside the unit circle), so $\oint \sec z\,dz = 0$.

GATE 2025 | Q.53 | 2 marks

Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential equations governing the system are given by

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t).$$

Which of the following statements is/are TRUE?

(A) The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs
(B) There exists a bounded input such that at least one of the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ is unbounded
(C) There exists a bounded input such that the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded
(D) The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded for all bounded inputs

Answer: (B) [self-solved, no official key available]

Solution. The state x_1 has eigenvalue $+2$ (unstable) and x_3 is an integrator, so a bounded input such as $u = 1$ gives unbounded states; but x_2 (eigenvalue -2) stays bounded, so not all three are unbounded.

GATE 2025 | Q.54 | 2 marks

The random variable X takes values in $\{-1, 0, 1\}$ with probabilities $P(X = -1) = P(X = 1) = \alpha$ and $P(X = 0) = 1 - 2\alpha$, where $0 < \alpha < \frac{1}{2}$. Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α. Which of the following statements is/are TRUE?

(A) $g(0.4) > g(0.3)$
(B) $g(0.3) > g(0.4)$
(C) $g(0.3) > g(0.25)$
(D) $g(0.25) > g(0.3)$

Answer: (B), (C) **(multiple select)** [self-solved, no official key available]

Solution. $g(\alpha) = -2\alpha \log_2 \alpha - (1 - 2\alpha) \log_2(1 - 2\alpha)$ has its maximum at $\alpha = 1/3$. $g(0.3) = 1.571$, $g(0.4) = 1.522$, $g(0.25) = 1.5$, so $g(0.3) > g(0.4)$ and $g(0.3) > g(0.25)$.

GATE 2025 | Q.55 | 2 marks

Let $f(t)$ be a periodic signal with fundamental period $T_0 > 0$. Consider the signal $y(t) = f(\alpha t)$, where $\alpha > 1$. The Fourier series expansions of $f(t)$ and $y(t)$ are given by $f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\frac{2\pi}{T_0}kt}$ and $y(t) = \sum_{k=-\infty}^{\infty} d_k e^{j\frac{2\pi}{T_0}\alpha kt}$. Which of the following statements is/are TRUE?

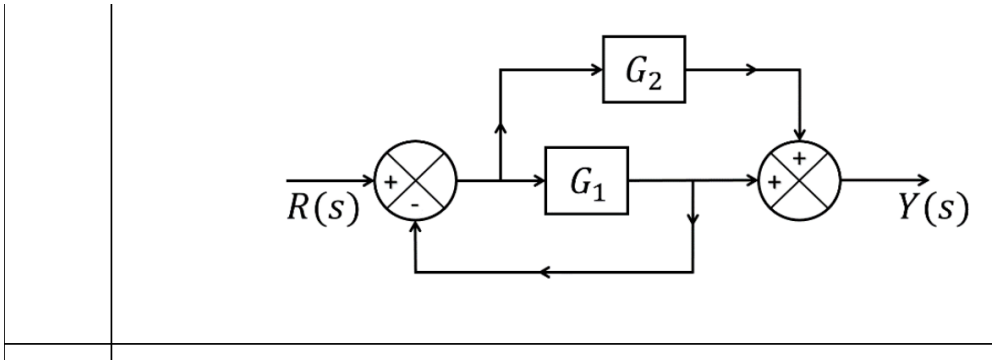
(A) $c_k = d_k$ for all k
(B) $y(t)$ is periodic with a fundamental period αT_0
(C) $c_k = d_k/\alpha$ for all k
(D) $y(t)$ is periodic with a fundamental period T_0/α

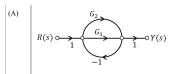
Answer: (A), (D) **(multiple select)** [self-solved, no official key available]

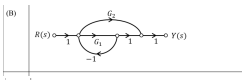
Solution. $y(t) = f(\alpha t) = \sum c_k e^{j\frac{2\pi}{T_0}k\alpha t}$, so $d_k = c_k$; the period shrinks to T_0/α.

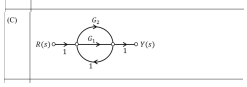
GATE 2025 | Q.56 | 2 marks

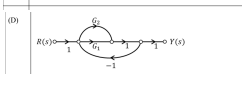
Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).



(A)


(B)


(C)


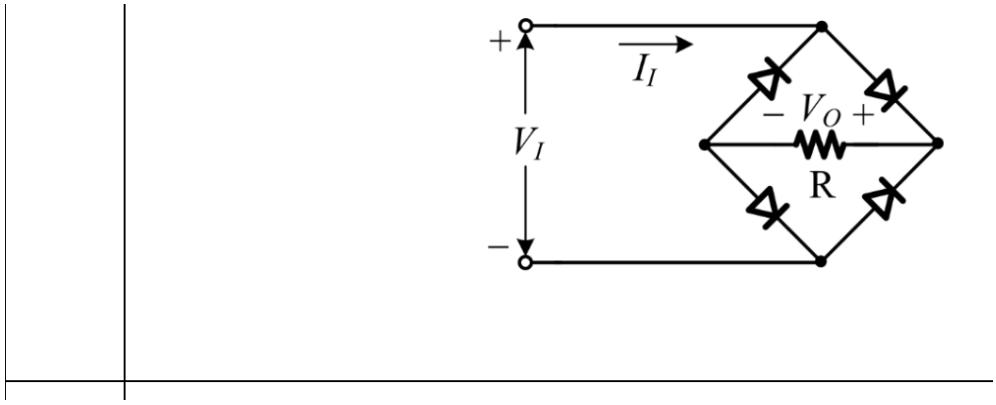
(D)


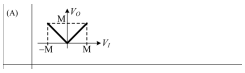
Answer: (A), (B) **(multiple select)** [self-solved, no official key available]

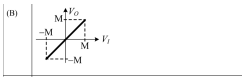
Solution. From the block diagram $Y = (G_1 + G_2)(R - Y)$: two parallel forward paths (G_1, G_2) from the error node to the output with unity negative feedback. Graphs (A) and (B) show exactly this structure.

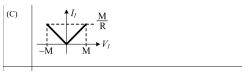
GATE 2025 | Q.57 | 2 marks

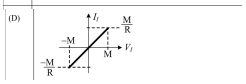
All the diodes in the circuit given below are ideal. Which of the following plots is/are correct when V_I (in Volts) is swept from $-M$ to M?



(A) 

(B) 

(C) 

(D) 

Answer: (A), (D) (multiple select) [self-solved, no official key available]

Solution. The bridge rectifier gives $V_O = |V_I|$ (plot A), while the source current is $I_I = V_I/R$, a signed linear function (plot D).

GATE 2025 | Q.58 | 2 marks

Two fair dice (with faces labeled $1, 2, 3, 4, 5$, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained. The expectation of X is _____ (rounded off to two decimal places).

Answer: 7.00 [self-solved, no official key available]

Solution. $E[X] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7$.

GATE 2025 | Q.59 | 2 marks

Consider the vectors $\mathbf{a} = \begin{bmatrix}1\\1\end{bmatrix}$, $\mathbf{b} = \begin{bmatrix}0\\3\sqrt{2}\end{bmatrix}$. For real-valued scalar variable x, the value of $\min_x \|\mathbf{a}x - \mathbf{b}\|_2$ is _____ (rounded off to two decimal places). $\|\cdot\|_2$ denotes the Euclidean norm, i.e., for $\mathbf{y} = \begin{bmatrix}y_1\\y_2\end{bmatrix}$, $\|\mathbf{y}\|_2 = \sqrt{y_1^2 + y_2^2}$.

Answer: 3.00 [self-solved, no official key available]

Solution. The minimising $x = \dfrac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}} = \dfrac{3\sqrt{2}}{2}$; the residual is $\mathbf{b} - x\mathbf{a} = \left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$ with length $\sqrt{9/2 + 9/2} = 3$.

GATE 2025 | Q.60 | 2 marks

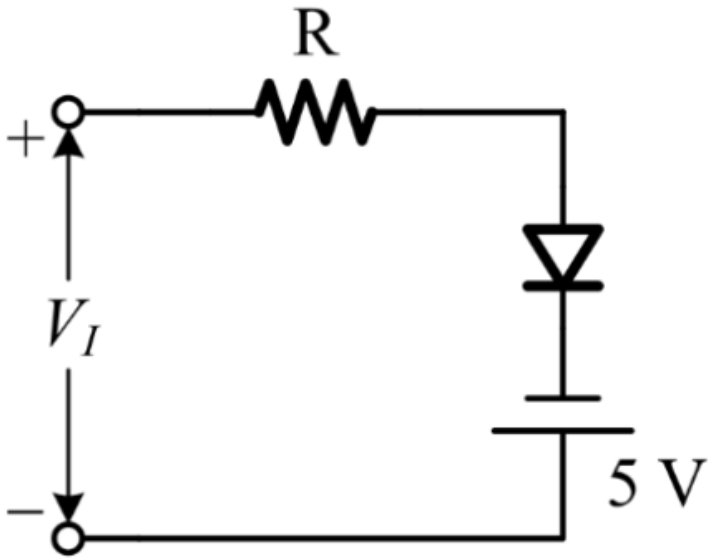
X and Y are Bernoulli random variables taking values in $\{0, 1\}$. The joint probability mass function of the random variables is given by: $P(X = 0, Y = 0) = 0.06$, $P(X = 0, Y = 1) = 0.14$, $P(X = 1, Y = 0) = 0.24$, $P(X = 1, Y = 1) = 0.56$. The mutual information $I(X; Y)$ is _____ (rounded off to two decimal places).

Answer: 0.00 [self-solved, no official key available]

Solution. The marginals are $P(X=0)=0.2$, $P(Y=0)=0.3$ and $P(x,y)=P(x)P(y)$ in all four cells (e.g. $0.2\times0.3=0.06$). So X, Y are independent and $I(X;Y)=0$.

GATE 2025 | Q.61 | 2 marks

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $V_I = 10\sin 100\pi t$, where time t is in seconds. The time duration (in ms, rounded off to two decimal places) for which the diode is forward biased during one period of the input is _____.

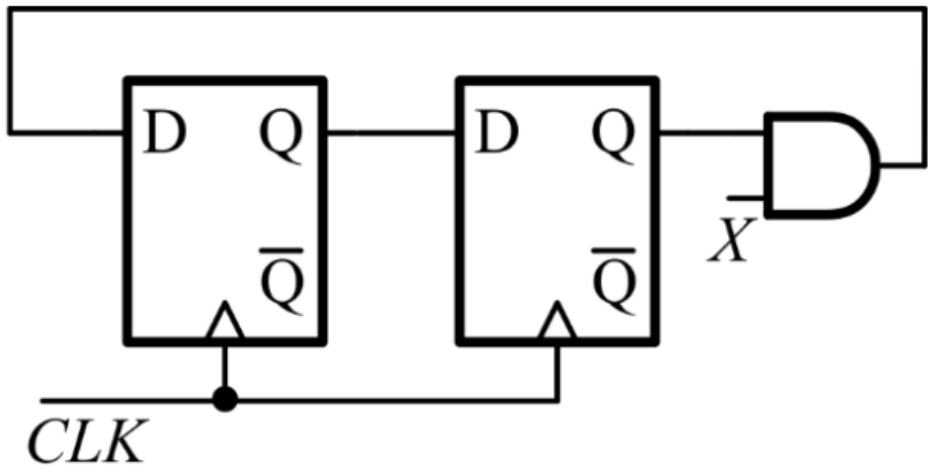


Answer: 6.67 [self-solved, no official key available]

Solution. The diode conducts when $V_I > 5$ V: $10\sin\theta > 5 \Rightarrow \theta \in (\pi/6, 5\pi/6)$, i.e. one third of the 20 ms period: 6.67 ms.

GATE 2025 | Q.62 | 2 marks

In the circuit shown below, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have a set-up time of 2 ns, a hold-time of 0 ns, and a clock-to-Q delay of 2 ns. The maximum clock frequency (in MHz, rounded off to the nearest integer) such that there are no setup violations is _____.



Answer: 200 [self-solved, no official key available]

Solution. Critical path $= t_{cq} + t_{AND} + t_{setup} = 2 + 1 + 2 = 5$ ns, hence $f_{max} = 1/5$ ns $= 200$ MHz.

GATE 2025 | Q.63 | 2 marks

An ideal p-n junction germanium diode has a reverse saturation current of 10 μA at 300 K. The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____. (Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J/K and the charge of an electron $e = 1.6 \times 10^{-19}$ C.)

Answer: 0.24 [self-solved, no official key available]

Solution. $V = V_T \ln\left(\frac{I}{I_s} + 1\right) = 0.02588 \ln(10001) = 0.238 \approx 0.24$ V.

GATE 2025 | Q.64 | 2 marks

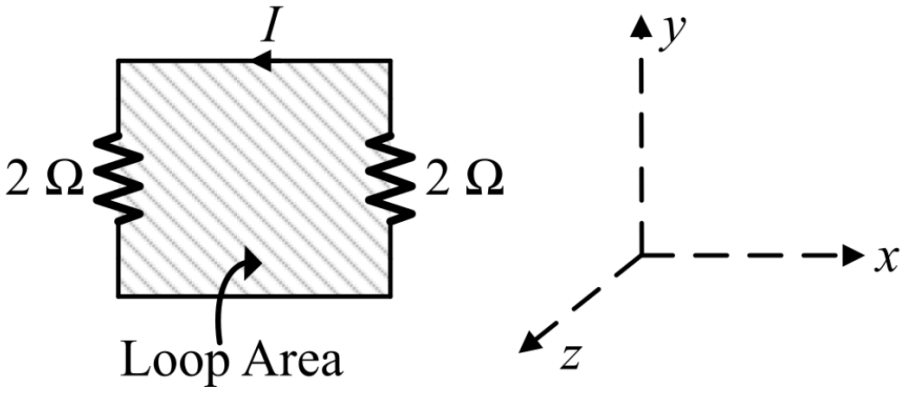
A 50 Ω lossless transmission line is terminated with a load Z_L of $(50 - j75)$ Ω. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

Answer: 6.4 [self-solved, no official key available]

Solution. $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-j75}{100 - j75}$, $|\Gamma|^2 = \frac{75^2}{100^2 + 75^2} = 0.36$. Delivered power $= P_{inc}(1 - |\Gamma|^2) = 10 \times 0.64 = 6.4$ mW.

GATE 2025 | Q.65 | 2 marks

Two resistors are connected in a circuit loop of area 5 m^2, as shown in the figure below. The circuit loop is placed on the x-y plane. When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z-axis, the magnitude of current I in the loop (in Amperes, rounded off to two decimal places) is _____.



Answer: 0.63 [self-solved, no official key available]

Solution. emf $= A\frac{dB}{dt} = 5 \times 0.5 = 2.5$ V; the loop resistance is $2 + 2 = 4$ Ω, so $I = 2.5/4 = 0.625 \approx 0.63$ A.